

Physical Chemistry of Materials

Lecture 24

기말고사 안내 및 복습



기말고사 안내

- 시간: 12월 19일(목) 오전 10:30~11:30
- 장소: 화학관 414호(홀수 학번), 114호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

액체와 유리(18강)

고분자의 단일 사슬 구조(19강)

실제 사슬 / 고분자 용액(20강)

가시화와 젤화(21강)

고분자의 점탄성(22강)

단백질(23강)

시험 문제

총 일곱 문제

개념을 묻는 문제: 세 문제

공식을 유도하는 문제: 세 문제

보너스 문제: 한 문제

소재물리화학의 구성

고전 열역학

열역학 복습

상 전이

통계역학

기초 이론

이상 기체

상호작용 계

고체

고체의 구조

고체의 진동

고체의 전기적 성질

고체의 자기적 성질

표면화학

~~나노화학~~ 응용

연성물질

액체

유리

고분자의 구조

고분자의 상 거동

고분자의 젤화

액정, 콜로이드, ...

소재물리화학의 구성

고전 열역학

열역학 복습

상 전이

통계역학

기초 이론

이상 기체

상호작용 계

고체

고체의 구조

고체의 진동

고체의 전기적 성질

고체의 자기적 성질

표면화학

~~나노화학~~ 응용

연성물질

액체 및 유리

고분자의 구조

고분자 용액

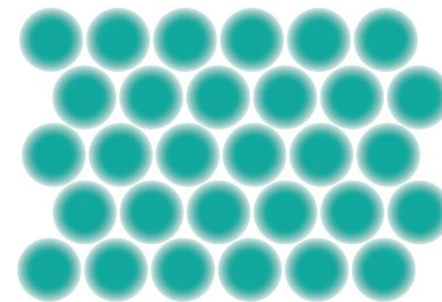
고분자 네트워크

고분자의 점탄성

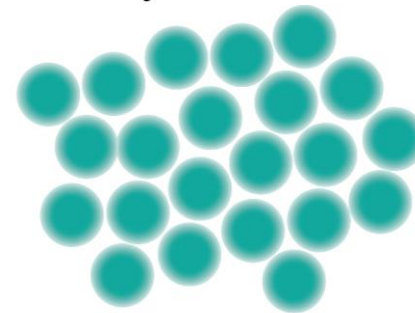
생체고분자

Condensed Phases

- Crystalline solids
 - Amorphous solids
 - Liquids (cf. glasses)
- } ?
- “Soft matters”
 - Polymers
 - Colloids
 - Liquid crystals ...



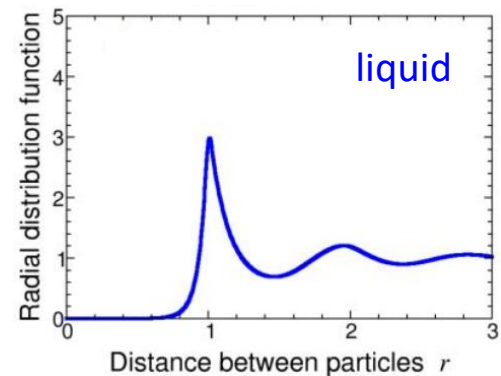
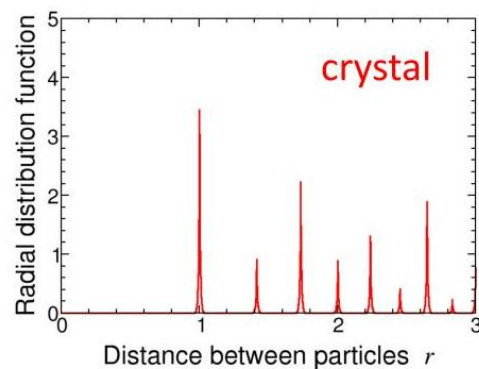
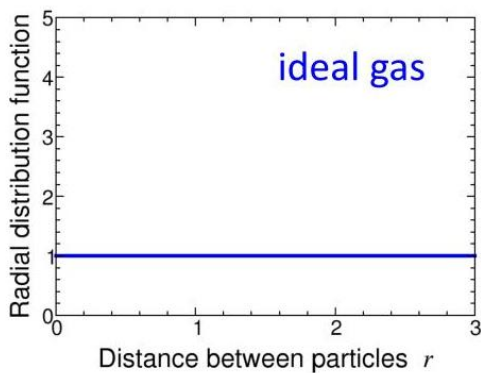
Crystalline



Amorphous

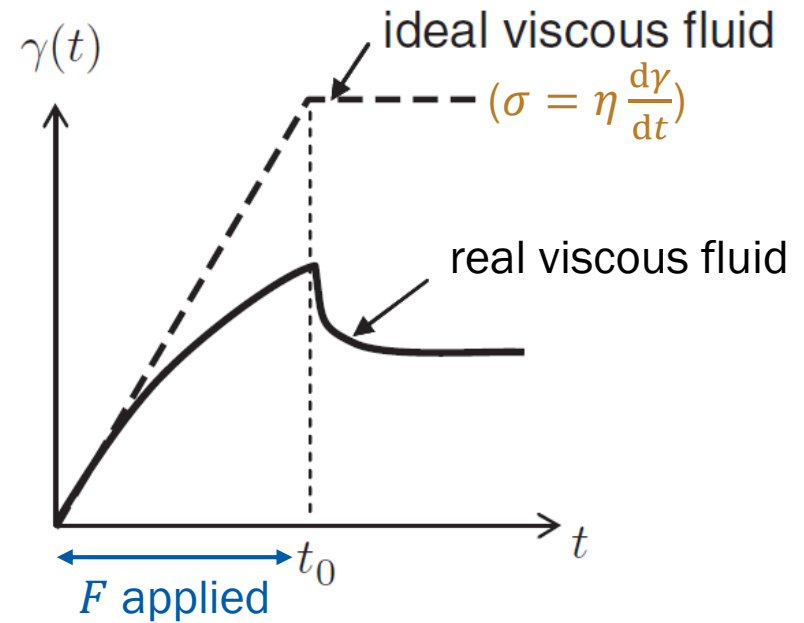
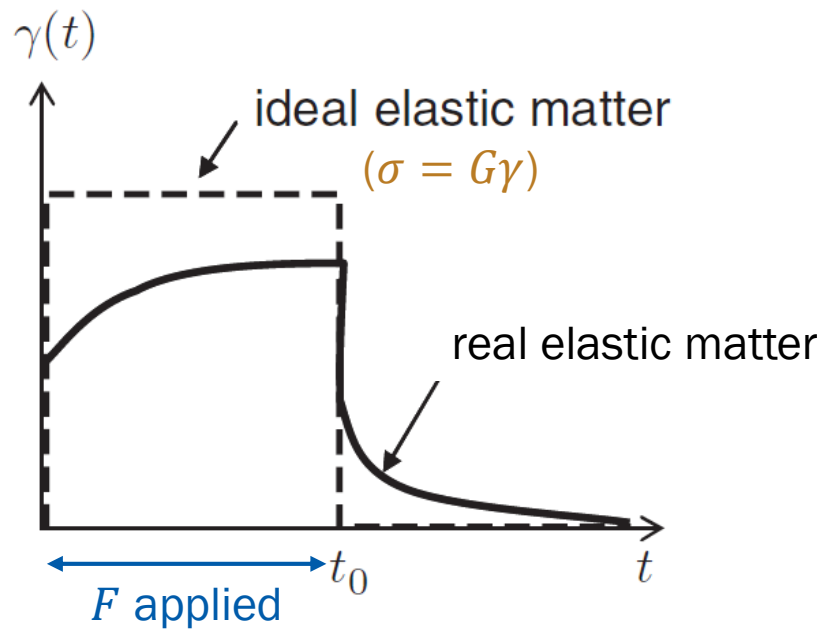
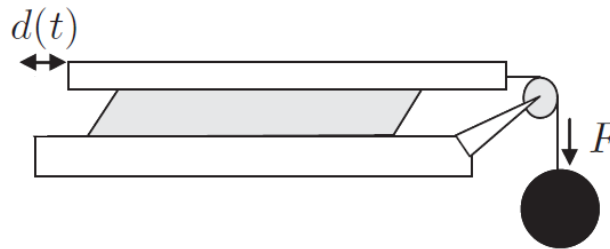
Amorphous Solids, Liquids, Glasses

- **Liquid-like structure:** radial distribution function



- **Amorphous solids vs. liquids:** fluidity
- **Liquids vs. glasses:** relaxation time

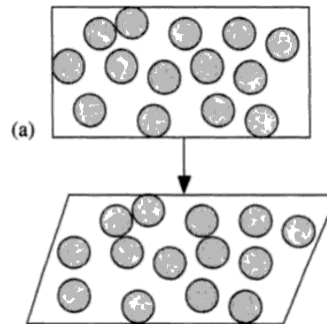
Elasticity and Viscosity



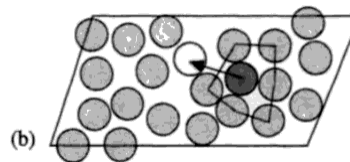
(Image: Doi, *Soft Matter Physics*, Fig. 3.2)

Molecular View on Viscosity

- Instantaneous response of a liquid: no rearrangement



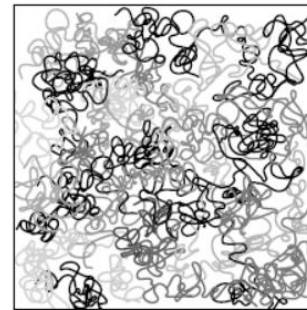
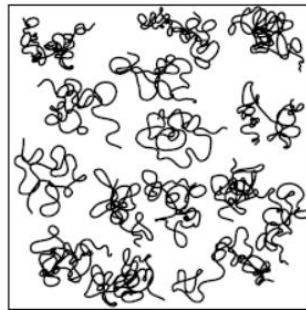
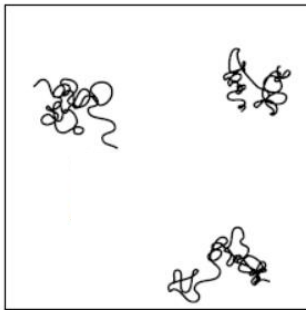
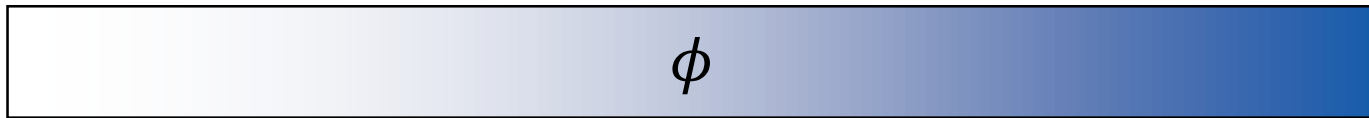
- Relaxation: escape from a “cage” by neighbors



Polymer Solution

0

1



↑
polymer melt

“overlap”

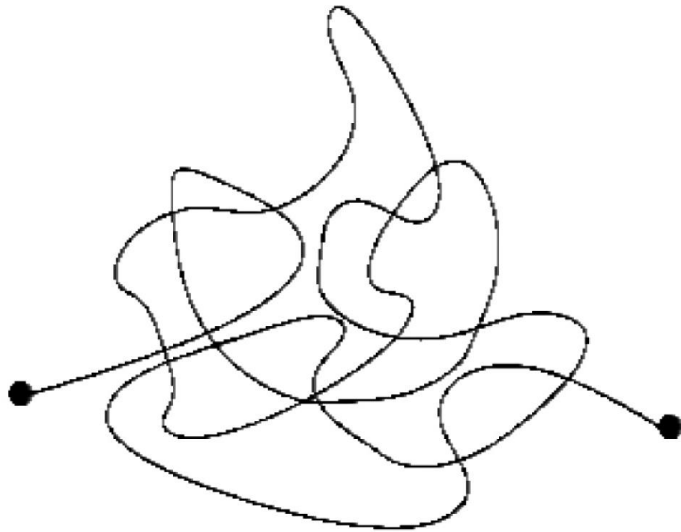
Lec 19, 20

Lec 20

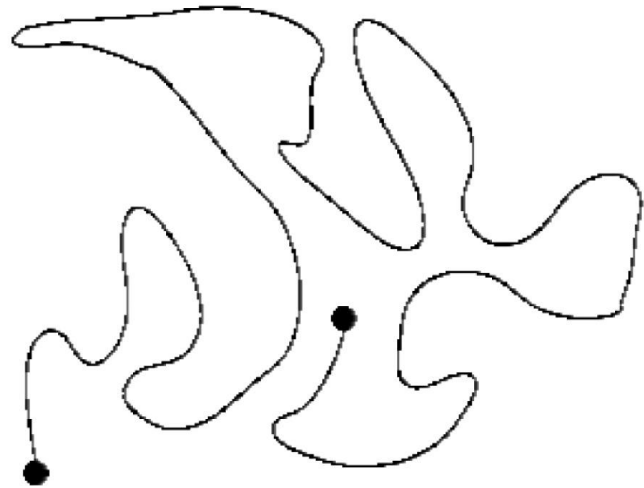
Lec 22

Single-Chain Structure

Ideal chains

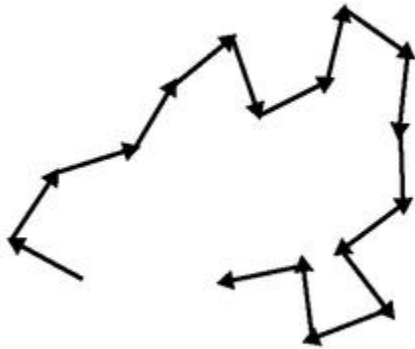


“Real” chains



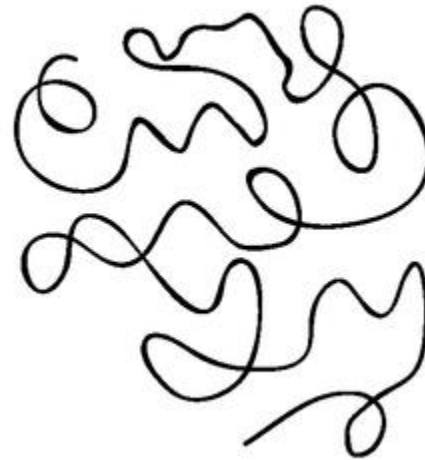
Models for Ideal Chains

A



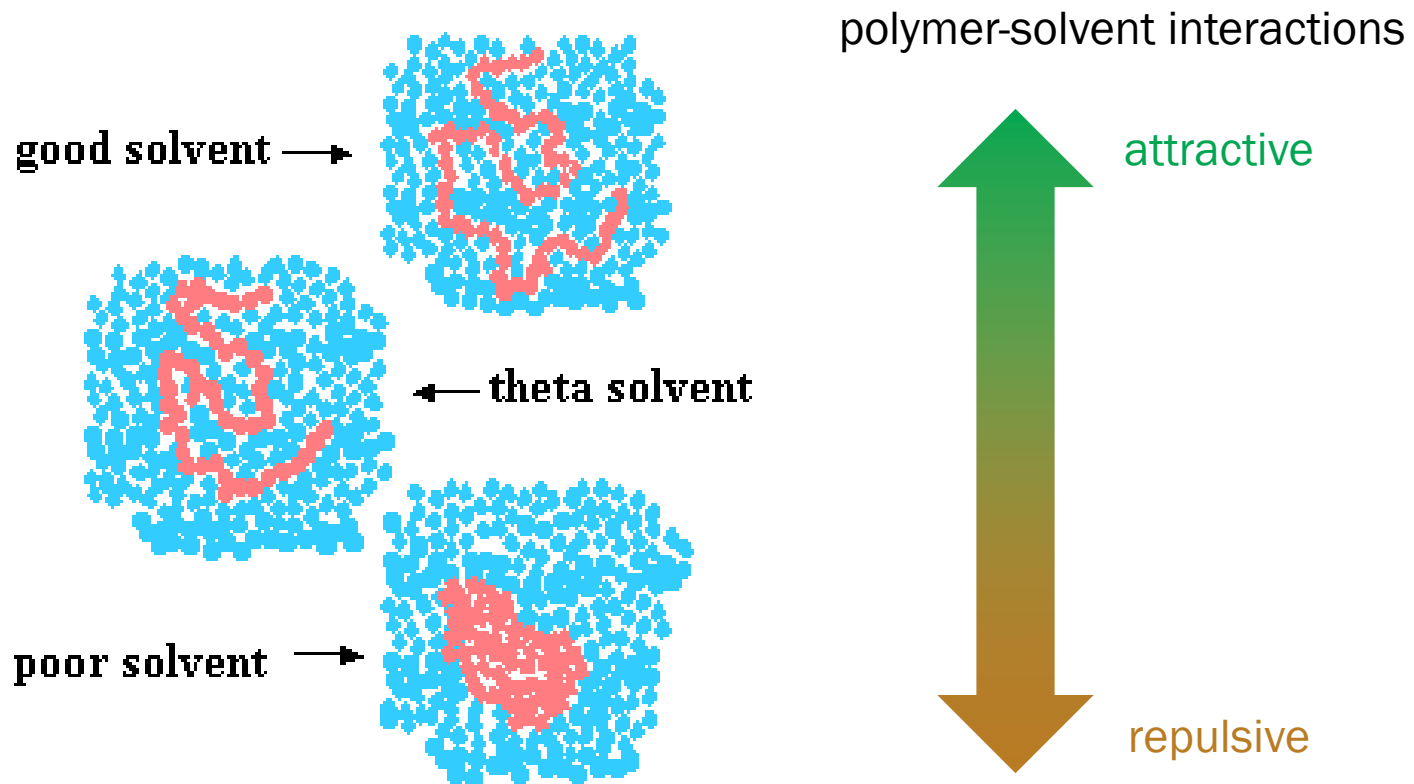
Freely jointed chain (FJC)

B



Worm-like chain (WLC)

Long-Range Interactions



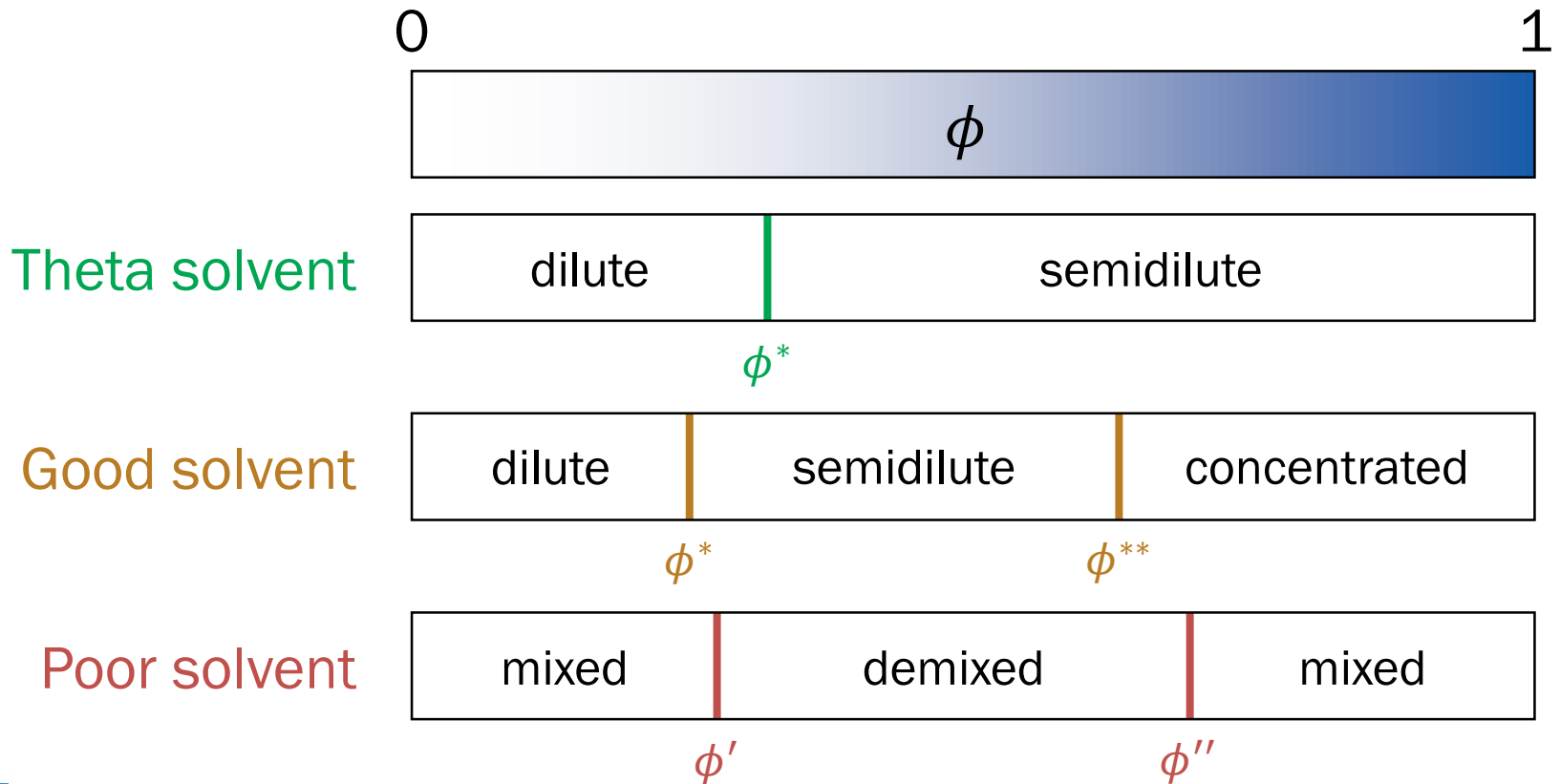
(Image: http://rkt.chem.ox.ac.uk/lectures/liqsolns/polymer_solutions.html)

Flory Exponent

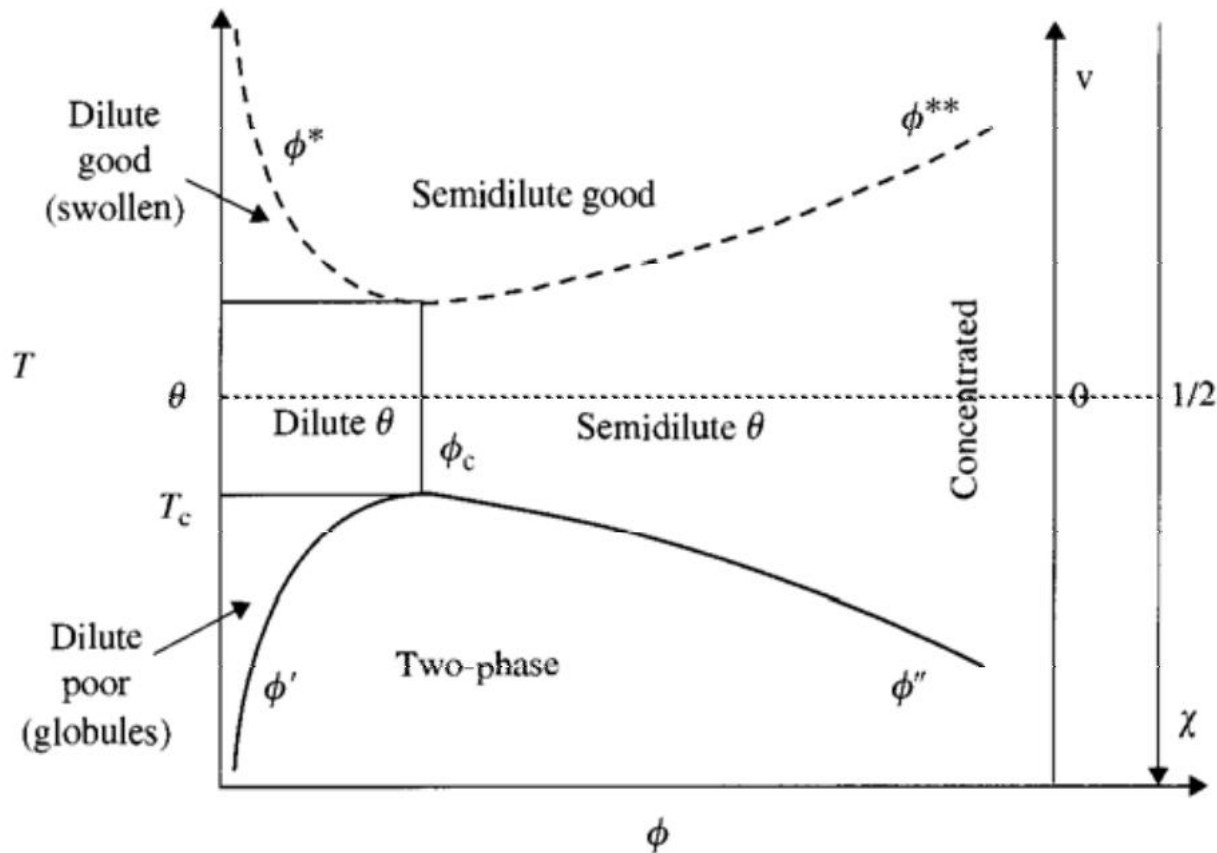
Generally, $R \propto N^\nu$

- Ideal chain (theta solvent): $\nu = 0.5$
- Good solvent: $\nu > 0.5$
 - Flory's prediction: $R_{\text{good}} = v_0^{1/5} a^{2/5} N^{3/5}$
- Poor solvent: $\nu < 0.5$
 - Flory's prediction: $R_{\text{poor}} = (w/|v_0|)^{1/3} N^{1/3}$

Concentration Dependence

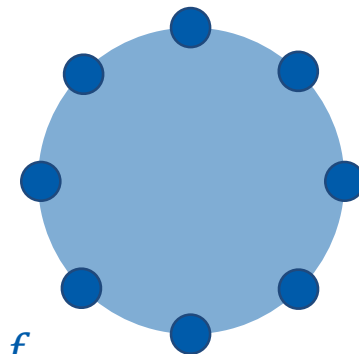
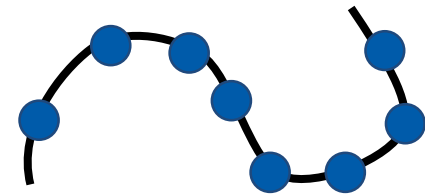
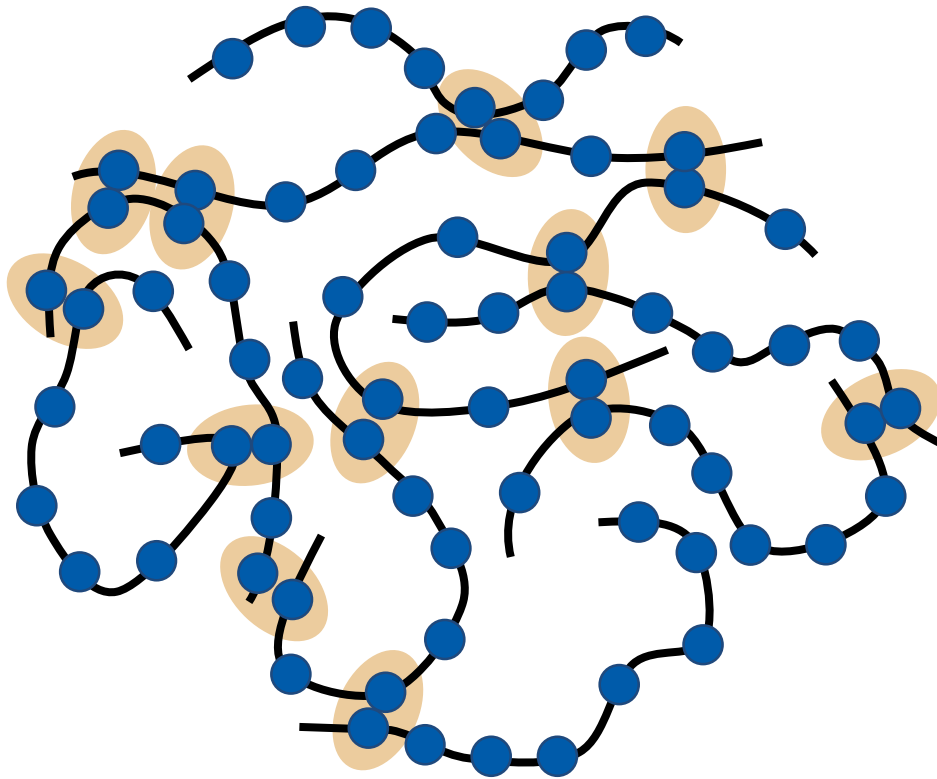


Phase Diagram of Polymer Solution



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 5.1)

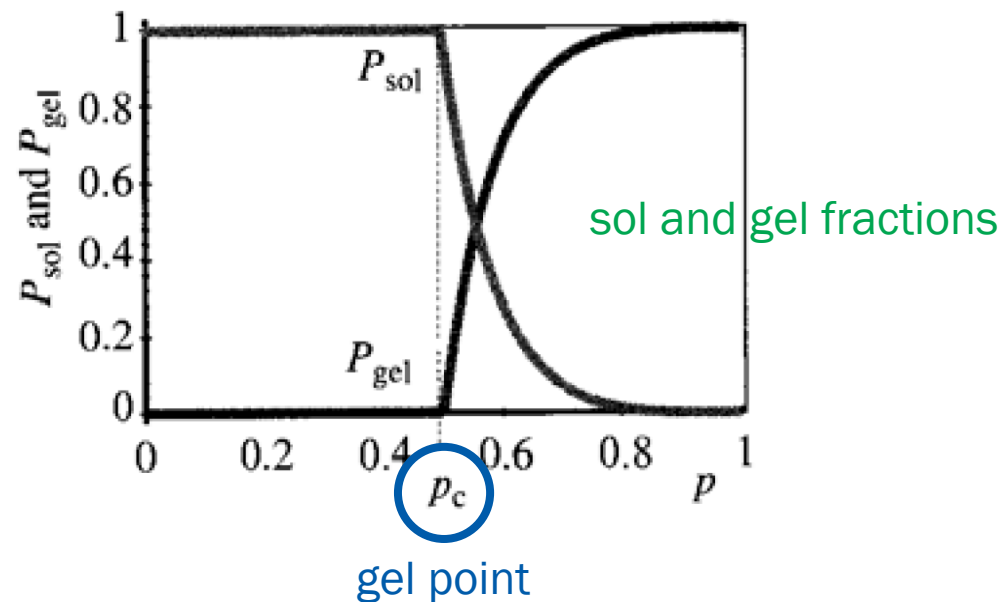
Gelation: Polymer Network



functionality f

Sol-Gel Transition

average cluster size
(= average number of
polymers per cluster)



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 6.15)

Flory-Stockmayer Theory

- Gel point

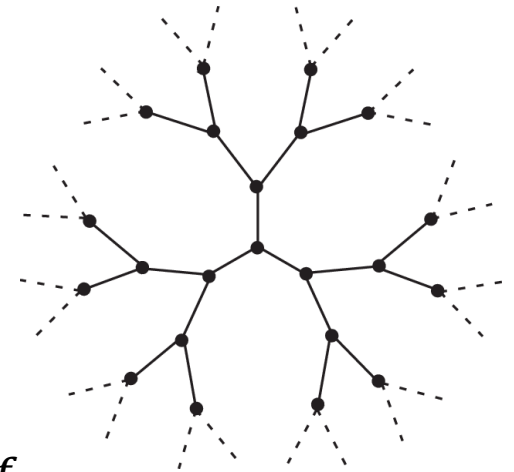
$$p_c = \frac{1}{f - 1}$$

- Sol fraction

$$P_{\text{sol}}^{1/f} = 1 - p + pP_{\text{sol}}^{(f-1)/f}$$

- Average number of polymers per cluster (for $p < p_c$)

$$N_n(p) = \frac{1}{1 - pf/2}$$



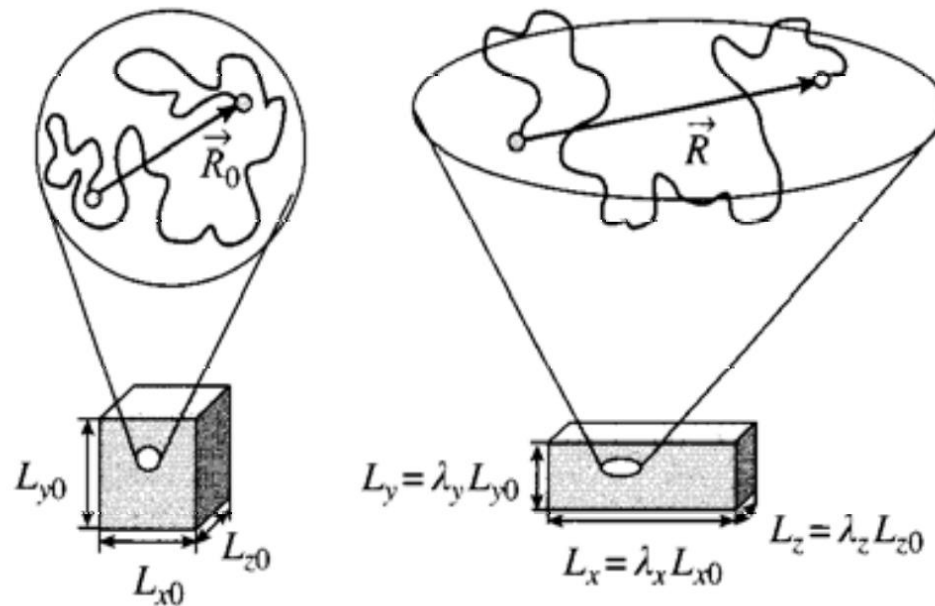
Rubber

- **Rubber:** polymer melt where crosslinks bond the chains together to form a macroscopic network



- “Liquid-like” state (temperature $> T_m$ or T_g)
- Bulk cannot flow due to crosslinks → “soft” solid

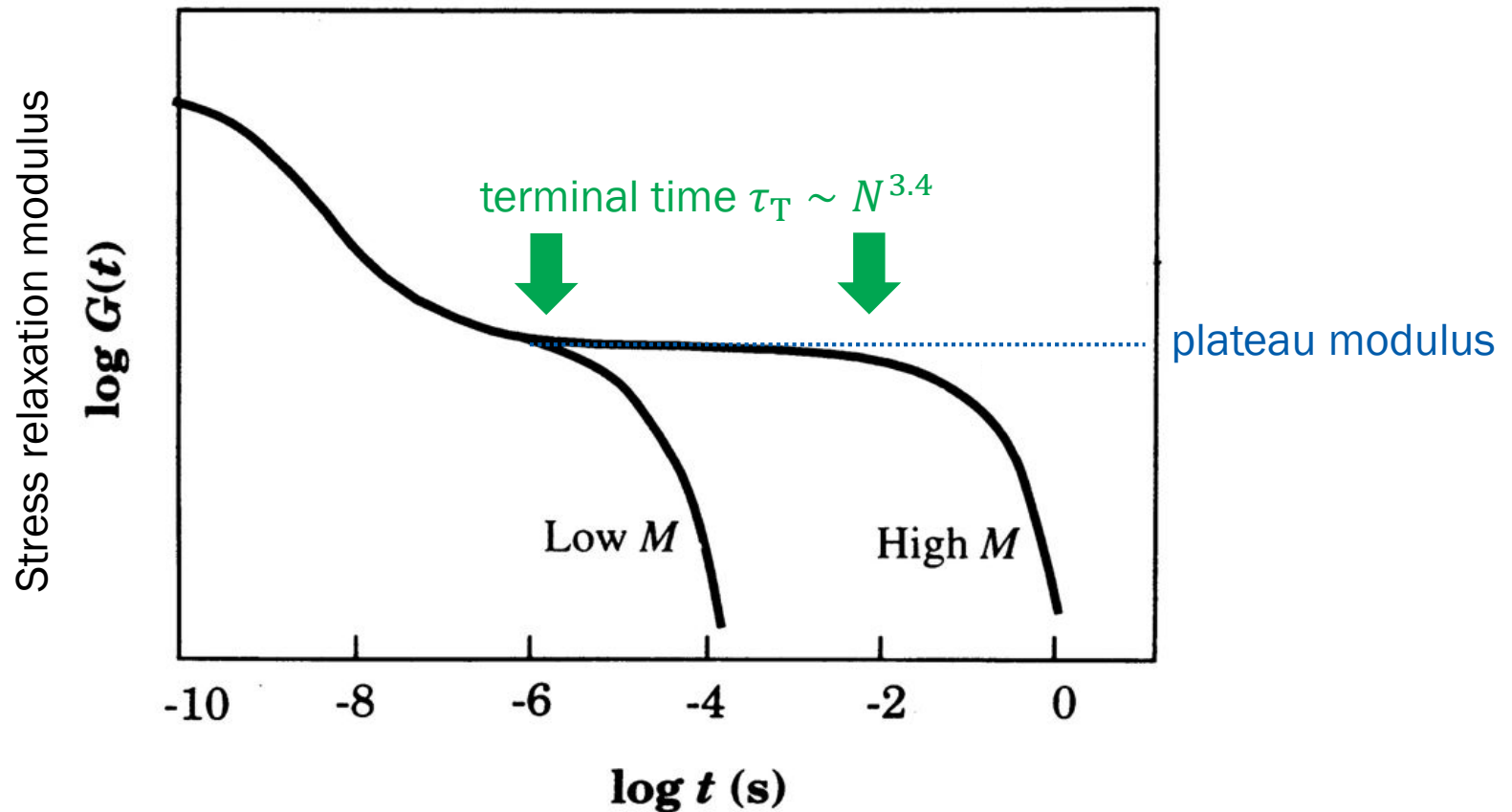
Affine Deformation



$$\Delta A_{\text{volume}} = \frac{nk_{\text{B}}T}{2} (\lambda_x^2 + \lambda_y^2 + \lambda_z^2 - 3)$$

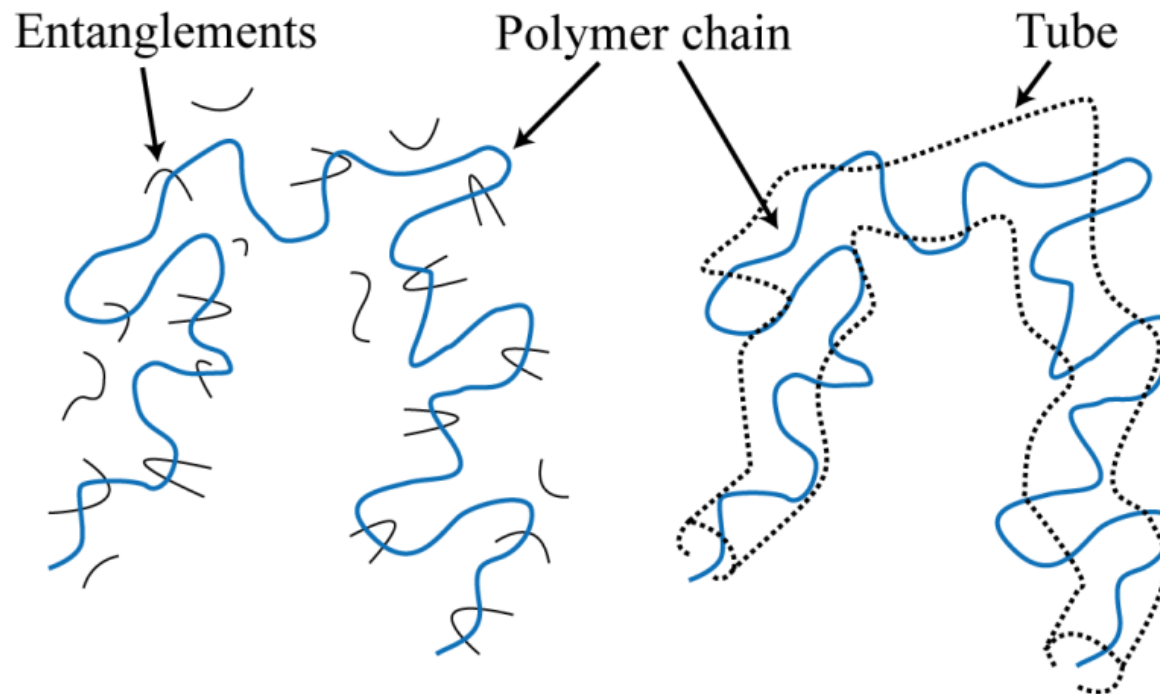
(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 7.1)

Viscoelasticity of Polymer Melts



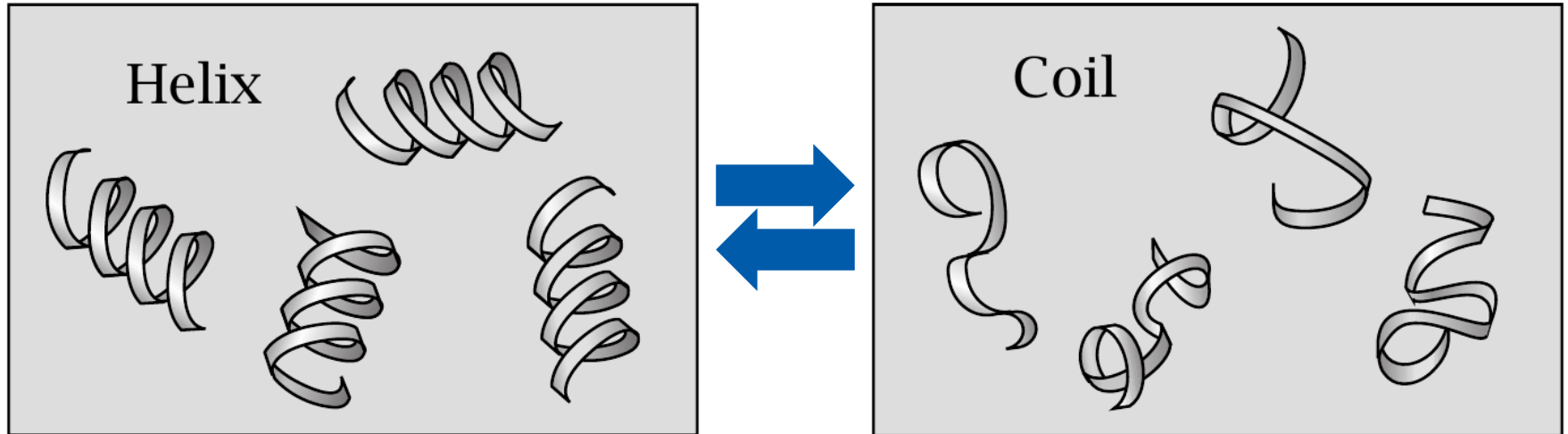
(Image: https://link.springer.com/chapter/10.1007/978-3-030-27701-7_20)

Viscosity: Tube Model



slithering motion: “reptation”

Helix-Coil Transition



Transition Models

- Independent monomer model

$$Z = z^N$$

- Zimm-Bragg model: cooperativity

$$Z = (1 \quad 1) \mathbf{T}^{N-1} \vec{z}_1$$

- Two-state model: maximum cooperativity

$$Z = e^{-\beta E(\text{CCC...C})} + e^{-\beta E(\text{HHH...H})}$$